From $10,788 in Losses to Data-Driven Edge:
A Case Study in Sports Betting Market Inefficiency and Applied Machine Learning

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Abstract

This paper presents a personal case study analyzing 4,699 verified sports bets totaling $37,392 in risk and $10,788 in losses over 17 months. Through rigorous post-hoc analysis, I identify the specific behavioral and mathematical errors that led to these losses, demonstrate the theoretical framework for finding exploitable edges in sports betting markets, and document my transition to a data-driven approach using machine learning and Monte Carlo simulations. This is not a guide to "beating the books"—it's an honest examination of what went wrong, the mathematics that explain why, and the systematic approach I'm now taking to find legitimate edge. All betting data is verified through JuiceReel.

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1 Introduction: The Honest Truth

I lost $10,787.53 sports betting. Not estimated—verified. Every single bet tracked through JuiceReel across 4,699 wagers over 17 months.

This paper serves three purposes:

1. **Accountability**: Document exactly what I did wrong, with data
2. **Education**: Explain the mathematics of why recreational bettors lose
3. **Framework**: Present my current approach to finding legitimate edge

I'm not selling picks. I'm not promoting a service. I'm a Master's student in Data Science using this problem as a real-world laboratory for applied machine learning. If you're betting recreationally and losing, this paper will show you exactly why—and what it actually takes to find edge.

2 Literature Review: Why You Can't "Beat the Books"

2.1 The Efficient Market Hypothesis in Sports Betting

Sports betting markets exhibit many characteristics of efficient financial markets [1]. The concept of market efficiency, originally developed by Fama (1970) for securities markets, applies directly: prices (odds) reflect all available information, making consistent outperformance theoretically impossible.

Key insight: Sportsbooks don't need to predict outcomes perfectly. They need to set lines that balance action and guarantee profit through the vigorish (vig).

2.2 The Vigorish: The Mathematical House Edge

The vig is the sportsbook's built-in commission. For standard -110/-110 lines:

$$\text{Implied Probability} = \frac{110}{110 + 100} = 52.38\% \tag{1}$$

Both sides imply 52.38% probability, totaling 104.76%—the extra 4.76% is the vig.

Breakeven requirement: To profit at -110 odds, you must win 52.38% of bets. Most recreational bettors win approximately 50%, guaranteeing long-term losses.

$$\text{Expected Value} = (0.50 \times 0.909) + (0.50 \times -1.00) = -0.0455 = -4.55\% \tag{2}$$

At a 50% win rate on -110 lines, you lose 4.55 cents per dollar wagered.

2.3 Market Efficiency vs. Exploitable Inefficiency

While betting markets are largely efficient, research has identified systematic inefficiencies:

- **Closing Line Value (CLV)**: Lines that move toward a bettor's position indicate the market agrees with their assessment—a proxy for +EV [2]
- **Opening Line Inefficiency**: Early lines contain more noise than closing lines
- **Player Prop Markets**: Less liquid than game lines, creating temporary mispricings
- **Cross-Market Arbitrage**: Different books price the same event differently

The critical distinction: You cannot consistently beat efficient markets through intuition or "feel." Edge comes from identifying and exploiting specific, quantifiable inefficiencies before the market corrects them.

2.4 Why Most Bettors Lose: Behavioral Economics

Beyond the mathematical house edge, behavioral biases compound losses:

- **Favorite-Longshot Bias**: Bettors overvalue longshots, making them -EV
- **Recency Bias**: Overweighting recent performance
- **Confirmation Bias**: Remembering wins, forgetting losses
- **Gambler's Fallacy**: Believing past outcomes affect future probability
- **Loss Chasing**: Increasing bet size after losses to "get even"

I exhibited all of these. The data proves it.

3 Methodology: Analyzing My Betting Data

3.1 Data Source and Verification

All betting data was tracked through JuiceReel, which syncs directly with sportsbook accounts. The dataset includes:

- 4,699 unique bets (13,368 individual legs)
- 12 sportsbooks
- 8 sports
- Date range: September 16, 2024 – February 2, 2026

3.2 Key Metrics Analyzed

Metric	**Definition**
ROI	(Profit/Loss) / Total Risked
Win Rate	Wins / (Wins + Losses)
CLV	(Closing Odds - Bet Odds) / Bet Odds
Expected Value	Win% × Payout - Loss% × Stake
Kelly Criterion	Optimal bet sizing given edge

Table 1: Key Performance Metrics

4 Results: The $10,788 Autopsy

4.1 Overall Performance

Metric	Value
Total Bets	4,699
Wins / Losses / Pushes	560 / 3,868 / 8
Win Rate	12.6%
Total Risked	$37,391.53
Net Profit/Loss	**-$10,787.53**
ROI	-28.85%

Table 2: Overall Betting Performance (Verified via JuiceReel)

A 12.6% win rate is catastrophically low. The explanation lies in bet structure.

Figure 1: Cumulative Profit/Loss Over Time. The steep decline from Sept-Dec 2024 shows the period of highest parlay activity and worst decision-making.

4.2 The Behavioral Pattern: What the Data Reveals About My Habits

Before analyzing *what* went wrong, it's important to understand *how* I was betting. The frequency and pattern data tells its own story:

Behavioral Metric	Value
Total days active	503 days
Average bets per day	9.3
Average dollars risked per day	$74.34
Average dollars risked per month	$2,230.11
Maximum bets in a single day	**186 bets**
Maximum risked in a single day	$1,490.00
Days with 10+ bets	140 days
Days with 20+ bets	78 days
Days with $100+ risked	100 days

Table 3: Betting Frequency Analysis: The Pattern of Compulsive Behavior

The story this tells: I wasn't casually betting—I was averaging nearly 10 bets per day, with 78 days where I placed 20+ bets. The maximum of 186 bets in a single day reveals compulsive behavior, not recreational entertainment.

The illusion of small bets: At $5-10 per bet, each wager felt insignificant. But $74/day compounds to $2,230/month—over $26,000 annually in betting volume. Small bets mask large exposure.

4.3 The Actual Dollar Breakdown by Bet Type

Legs	Bets	Total Risked	Total P/L	ROI
1 (Singles)	1,892	$12,386.67	-$3,037.40	-24.5%
2	853	$5,586.25	-$998.61	-17.9%
3	684	$5,045.30	-$2,671.38	-52.9%
4	551	$5,147.26	+$260.98	+5.1%
5	207	$2,061.15	-$1,007.63	-48.9%
6	207	$2,873.24	-$1,229.78	-42.8%
7+	305	$4,291.66	-$2,103.71	-49.0%
TOTAL	**4,699**	**$37,391.53**	**-$10,787.53**	**-28.8%**

Table 4: Actual Dollar Breakdown by Number of Legs

Key insight: 60% of my bets were parlays, but they accounted for 67% of total risk ($25,005 of $37,392). Parlays weren't just losing more often—I was betting more on them.

The parlay tax: If I had bet my entire $37,392 volume as singles at my actual singles ROI (-24.5%), I would have lost approximately $9,169. Instead, the parlay-heavy approach cost me $10,788—an extra $1,619 lost purely due to parlay structure.

4.4 What I Did Wrong #1: The Parlay Trap

Bet Type	Count	Net P/L	Win Rate	ROI
Singles (1 leg)	1,892	-$3,037.40	18.4%	-24.5%
Parlays (2+ legs)	2,807	-$7,750.13	8.5%	-31.0%

Table 5: Singles vs. Parlays: The Parlay Trap Cost Me $7,750

The math: For a 2-leg parlay at -110 per leg, you need both legs to win. If each leg has 52.38% implied probability:

$$P(\text{Both Win}) = 0.5238 \times 0.5238 = 27.44\% \tag{3}$$

But the actual breakeven is:

$$P(\text{Breakeven at } +260) = \frac{100}{260 + 100} = 27.78\% \tag{4}$$

The parlay offers +260, but breakeven requires 27.78%—the vig compounds. For a 6-leg parlay, this effect is devastating.

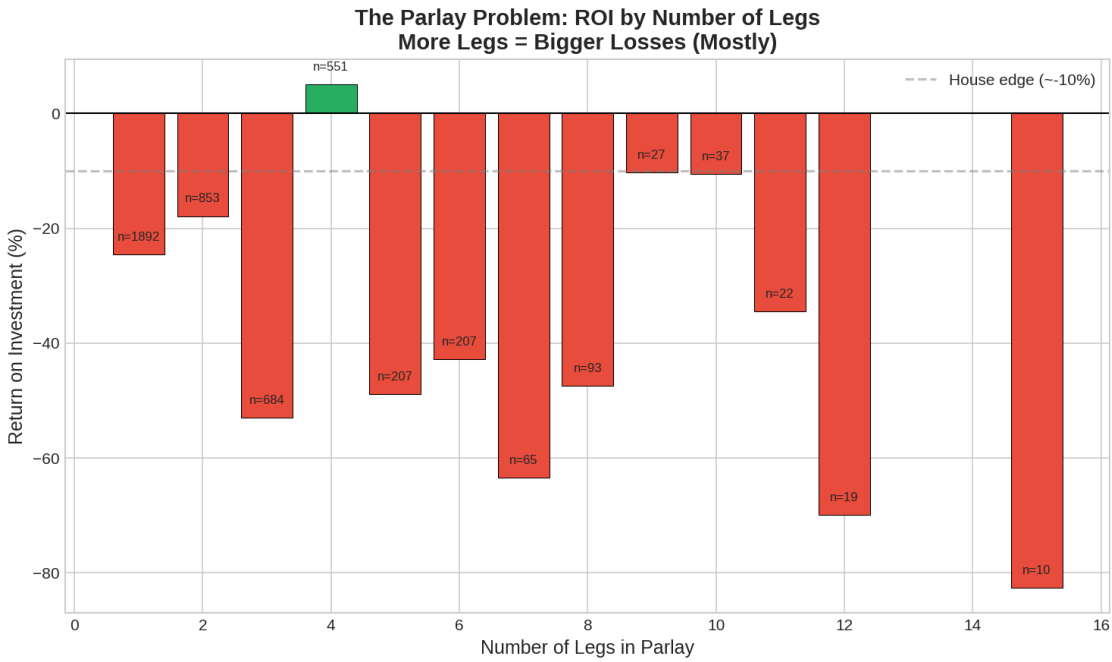

Figure 2: ROI by Number of Legs. Nearly every parlay size shows negative ROI, with larger parlays performing catastrophically. The lone exception (4-leg parlays at +5.1% ROI) is likely statistical noise.

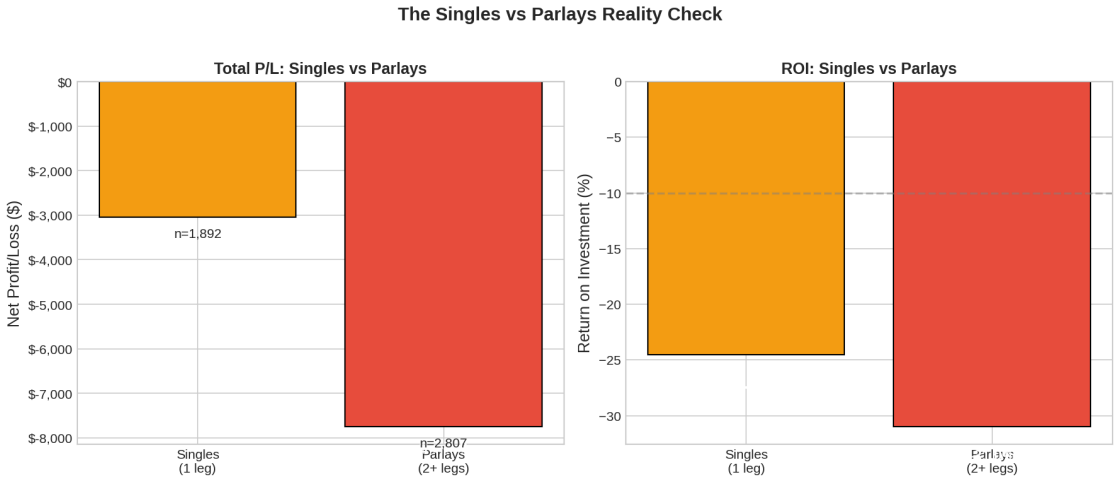

Figure 3: The Singles vs. Parlays Reality Check. Parlays lost $7,750 vs. $3,037 for singles, despite singles having fewer total bets.

4.5 What I Did Wrong #2: Sportsbook Selection

Sportsbook	Bets	Net P/L	ROI
FanDuel	1,608	-$4,362.35	-33.3%
DraftKings	1,913	-$3,999.92	-23.9%
Caesars	223	-$970.70	-68.9%
TheScoreBet	203	-$672.98	-44.3%
BetMGM	338	-$307.05	-14.7%
BetRivers	177	+$284.79	+33.9%
Fanatics	62	+$69.16	+16.9%

Table 6: Performance by Sportsbook. BetRivers and Fanatics showed profit—line shopping matters.

Lesson: FanDuel and DraftKings accounted for 77% of losses despite only 75% of volume. These are the most heavily traded books with the sharpest lines. BetRivers, a softer book, was profitable.

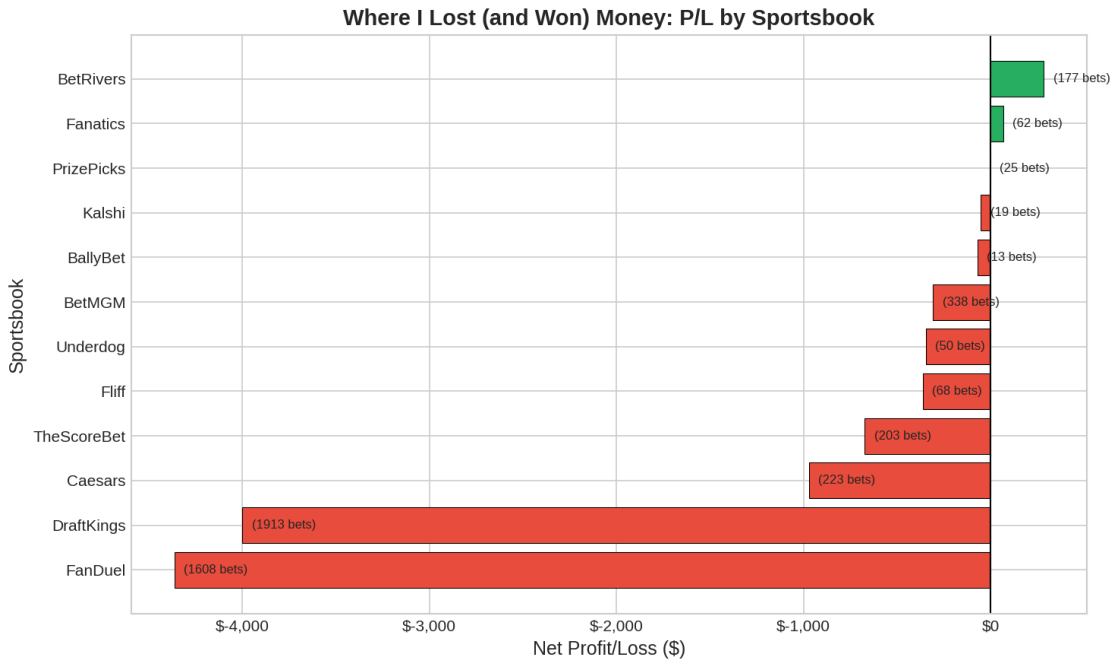

Figure 4: Net P/L by Sportsbook. The contrast between sharp books (FanDuel, DraftKings) and softer books (BetRivers) is stark.

4.6 What I Did Wrong #3: Sport Selection Without Edge

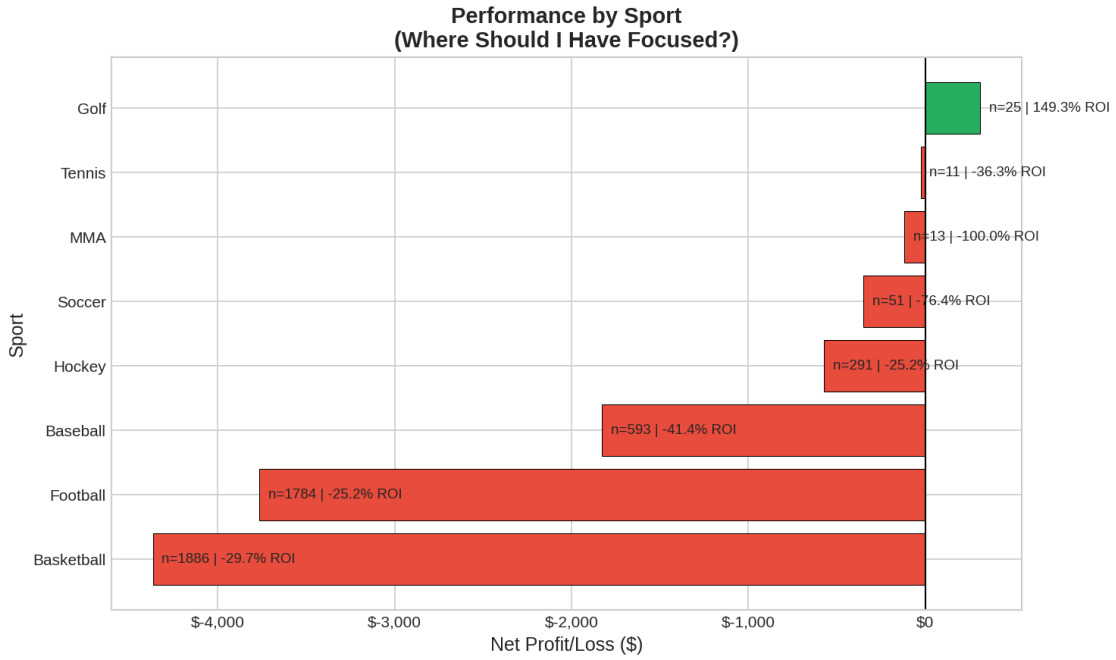

Figure 5: Performance by Sport. Golf was the only profitable sport (+149% ROI on 25 bets). Basketball and Football combined lost $8,131.

Sport	Bets	Net P/L	ROI
Basketball	1,886	-$4,367.64	-29.7%
Football	1,784	-$3,764.05	-25.2%
Baseball	593	-$1,828.30	-41.4%
Hockey	291	-$573.31	-25.2%
Soccer	51	-$348.81	-76.4%
Golf	25	+$312.18	+149.3%

Table 7: Performance by Sport

The irony: I bet heavily on sports I "knew" (basketball, football) and barely touched golf—yet golf was the only profitable sport. Knowledge != edge.

4.7 What I Did Wrong #4: No Bankroll Management

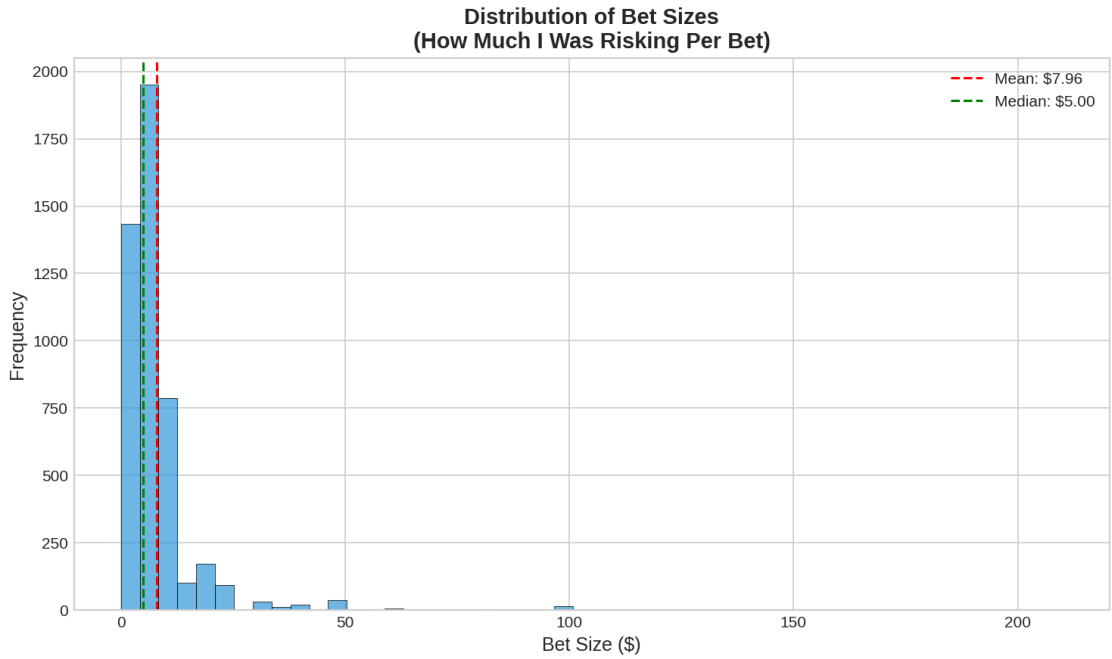

Figure 6: Distribution of Bet Sizes. Mean bet: $7.96, Median: $5.00. The inconsistency shows no structured approach to bet sizing.

Without Kelly Criterion or any systematic sizing, I was essentially guessing how much to risk on each bet. This compounds errors.

4.8 Monthly Performance: The Timeline of Destruction

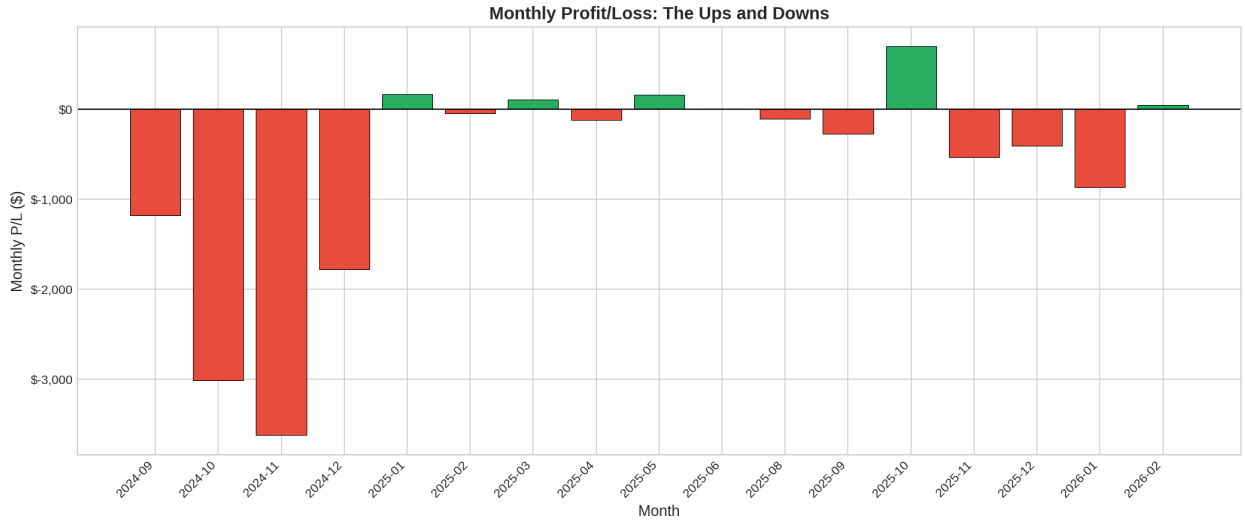

Figure 7: Monthly P/L. October-November 2024 saw the worst losses (-$6,500 combined), coinciding with NFL/NBA season overlap and maximum parlay activity.

5 My Biggest Regret: The CLV Data I Ignored

During a brief period, I tracked Closing Line Value on 87 bets:

Metric	Value
Bets with +CLV	79 (90.8%)
Bets with -CLV	8 (9.2%)
Average CLV	+0.02%
+CLV Bets Net Result	-$386.07
-CLV Bets Net Result	-$177.00

Table 8: CLV Analysis on 87 Tracked Bets

What this means: 90.8% of these bets beat the closing line, indicating I was finding +EV opportunities. But variance swung negative, I lost confidence, and I stopped tracking.

The statistical reality: 87 bets is far below the 500+ needed for statistical significance. I abandoned a potentially profitable approach before the sample could prove itself.

6 The Turning Point: Applying Data Science

6.1 The Framework: From Gut to Algorithm

After starting my Master's in Data Science, I rebuilt my approach from first principles:

1. **Data Pipeline**: 65+ BigQuery tables processing 100K+ data points daily

2. **Feature Engineering**: 25+ features per prop type (rolling averages, matchup factors, advanced metrics)
3. **Monte Carlo Simulations**: 10,000 simulations per prop to generate probability distributions
4. **Edge Calculation**: Compare model probability to implied odds for EV
5. **Kelly Criterion Sizing**: Size bets proportional to edge, not confidence

6.2 The Tool I Built

```
🎲 Monte Carlo Simulation for NBA Props
============================================================
Running 10,000 simulations per prop...

✅ Found 4 props to simulate

📈 Simulation Results:
------------------------------------------------------------
Brook Lopez           PTS O5.5   | Win:  68.3% | EV/100:  $+39.43 ✅
Thomas Bryant         PTS O7.5   | Win:  68.2% | EV/100:  $+39.05 ✅
Jonas Valanciunas     PTS O8.5   | Win:  62.7% | EV/100:  $+25.42 ✅
John Collins          REB O5.5   | Win:  68.3% | EV/100:  $+49.00 ✅
------------------------------------------------------------
```

Figure 8: Monte Carlo Simulation CLI Tool. Runs 10,000 simulations per prop, calculates win probability and EV per $100 wagered.

6.3 How Monte Carlo Simulations Work

For each player prop, the model:

1. Generates a probability distribution based on:
   - Rolling averages (L5, L10, season)
   - Opponent defensive metrics
   - Usage rate, pace, rest days
   - Home/away splits
2. Runs 10,000 simulations sampling from this distribution
3. Calculates the percentage of simulations that clear the sportsbook line
4. Converts to Expected Value per $100 wagered

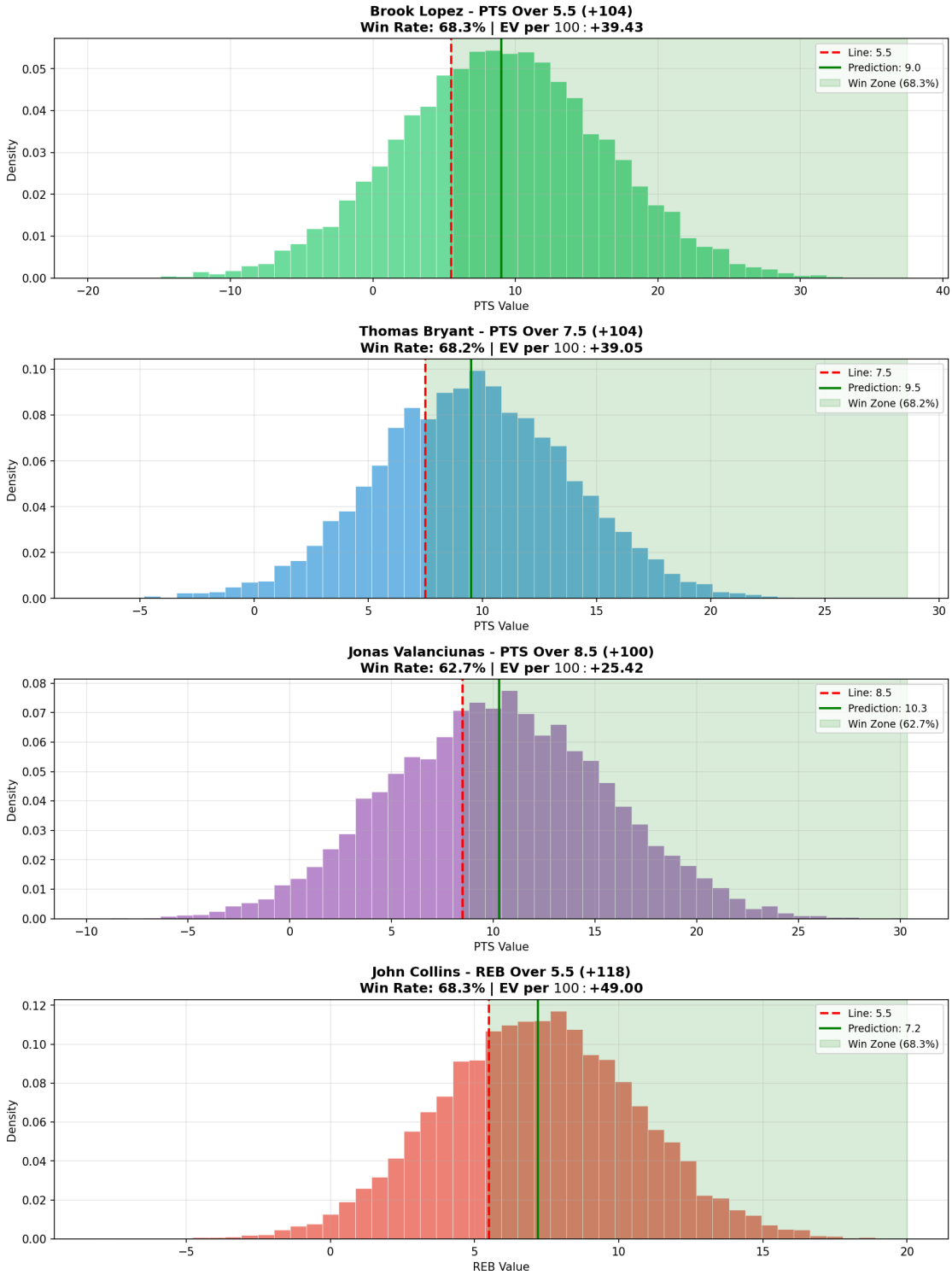

Figure 9: Monte Carlo Probability Distributions. Each chart shows the modeled distribution (histogram), the sportsbook line (red dashed), the model prediction (green line), and the win zone (shaded area representing probability of clearing the line).

6.4 Expected Value Calculation

For a +100 bet (even odds) with 68% win probability:

$$\text{EV} = (0.68 \times \$100) - (0.32 \times \$100) = \$68 - \$32 = +\$36 \text{ per } \$100 \tag{5}$$

This is the theoretical edge. Over hundreds of bets, this edge compounds.

6.5 Kelly Criterion for Bet Sizing

To avoid overbetting (what destroyed my bankroll), I use fractional Kelly:

$$f^* = \frac{p(b+1) - 1}{b} \tag{6}$$

Where:

- f^* = fraction of bankroll to bet
- p = probability of winning
- b = decimal odds - 1

I use quarter-Kelly ($0.25 \times f^*$) for conservative sizing.

7 Real Results: Proof of Concept

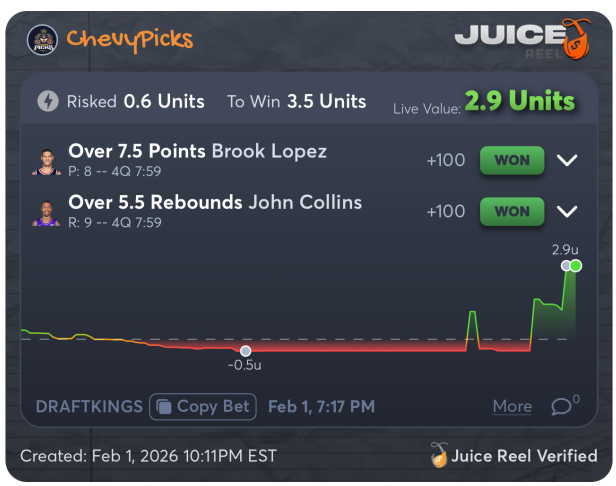

Figure 10: JuiceReel-verified win: 2-leg parlay at +2.9 units. Both props (Brook Lopez Over 7.5 PTS, John Collins Over 5.5 REB) were identified by the Monte Carlo model.

Important caveat: One win proves nothing. The model's value is in long-term expected value, not individual results. I'm sharing this as proof of concept, not proof of profitability.

8 What I'm Doing Differently Now

Before (Lost $10,788)	Now
Gut-feel betting	Model-driven selection
Heavy parlay focus	Primarily singles
Random bet sizing	Quarter-Kelly sizing
One or two books	Line shopping across 6+ books
Abandoned +CLV tracking	Continuous CLV monitoring
No variance planning	Prepared for 500+ bet sample

Table 9: Systematic Changes to Betting Approach

9 What You Can Learn From My Mistakes

9.1 If You're Betting Recreationally

1. **Accept the math**: Without edge, you will lose long-term. The vig guarantees it.
2. **Avoid parlays**: Each leg multiplies the house edge. That 10-leg parlay isn't a moonshot—it's a donation.
3. **Track everything**: You can't improve what you don't measure. Use JuiceReel or similar.
4. **Understand CLV**: If your bets consistently move against you after placement, you're being sharp-sided.

9.2 If You're Serious About Finding Edge

1. **Build infrastructure**: You need data pipelines, not spreadsheets
2. **Learn probability theory**: Monte Carlo, Bayesian inference, Kelly Criterion
3. **Plan for variance**: 500+ bet sample minimum before evaluating strategy
4. **Line shop aggressively**: The same bet at different books can be +EV or -EV
5. **Focus on inefficient markets**: Player props, early lines, soft books

10 Conclusion

I lost $10,787.53 sports betting. The data shows exactly why:

- 60% of bets were parlays, compounding the house edge
- No systematic approach to bet sizing or bankroll management
- Bet on sports I "knew" rather than where I had quantifiable edge
- Abandoned +EV tracking before reaching statistical significance

I'm not claiming my new approach will be profitable. Variance is real, and edge is never guaranteed. But I'm now betting with:

- Quantified expected value for every bet
- Systematic position sizing via Kelly Criterion
- Infrastructure to track and improve over time
- Realistic expectations about sample size and variance

This paper isn't about beating the books. It's about understanding why I lost, applying data science to find legitimate edge, and documenting the journey honestly.

All betting data verified via JuiceReel. Model performance based on backtesting—past results do not guarantee future returns. Gambling involves risk. Bet responsibly.

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[3] Kelly, J. L. (1956). A New Interpretation of Information Rate. *Bell System Technical Journal*, 35(4), 917-926.

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